

Lecture 5: Joint and Marginal Distributions, Conditioning, Independence

Jamie Haddock

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```
import pandas as pdd1 = rng.integers(1, 7, size=500000)d2 = rng.integers(1, 7, size=500000)X, Y = d1 +
```

::: {.callout-caution icon=false}## ExerciseAre the events $\{X = 12\}$ and $\{Y = 6\}$ independent?:::

Answer:

$P(X = 12) = 1/36$, $P(Y = 6) = 11/36$, and $P(X = 12, Y = 6) = 1/36$ (only (6,6) works). Since $1/36 \neq (1/36)(11/36)$, they are not independent – but note $P(X = 12 | Y = 6) = 1/11 \neq P(X = 12)$, consistent.

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